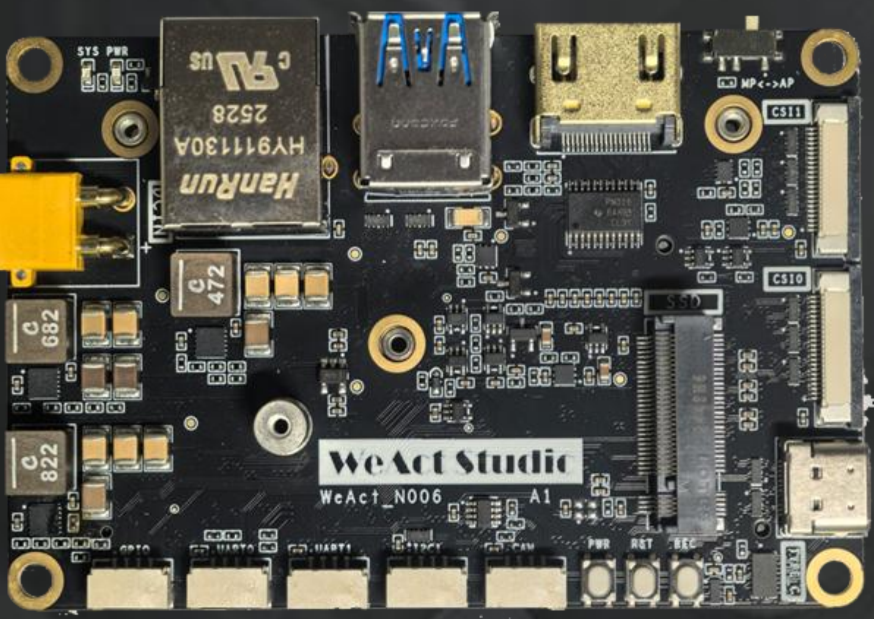


WeAct Studio

N006 CARRIER BOARD

Tutorial

WeAct Studio

Catalog

REVISION HISTORY

Draft Date	Revision	Description
2026.02.24	V1.0	1. Initial
2026.06.07	V1.1	1. Support Jetpack7.x

1. BUILD FLASH ENVIRONMENT

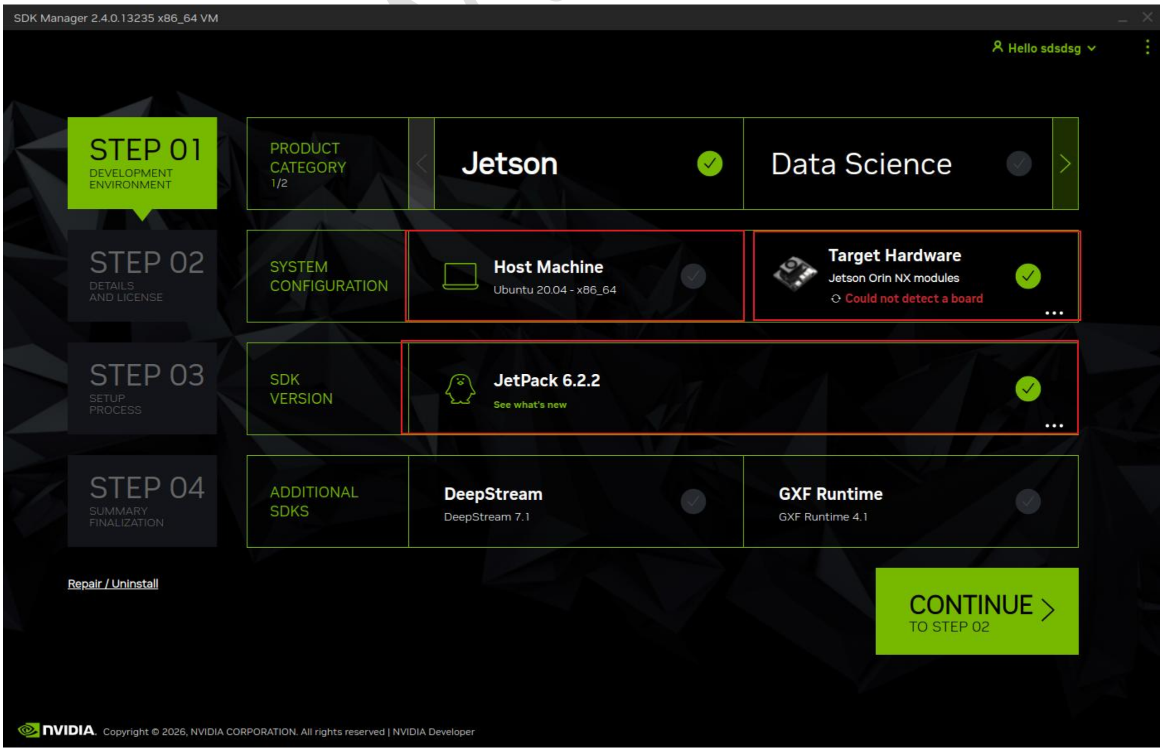
a) First, you need a computer with **Ubuntu 20.04** or above as the host to flash Orin NX, or you can install VMware on windows.

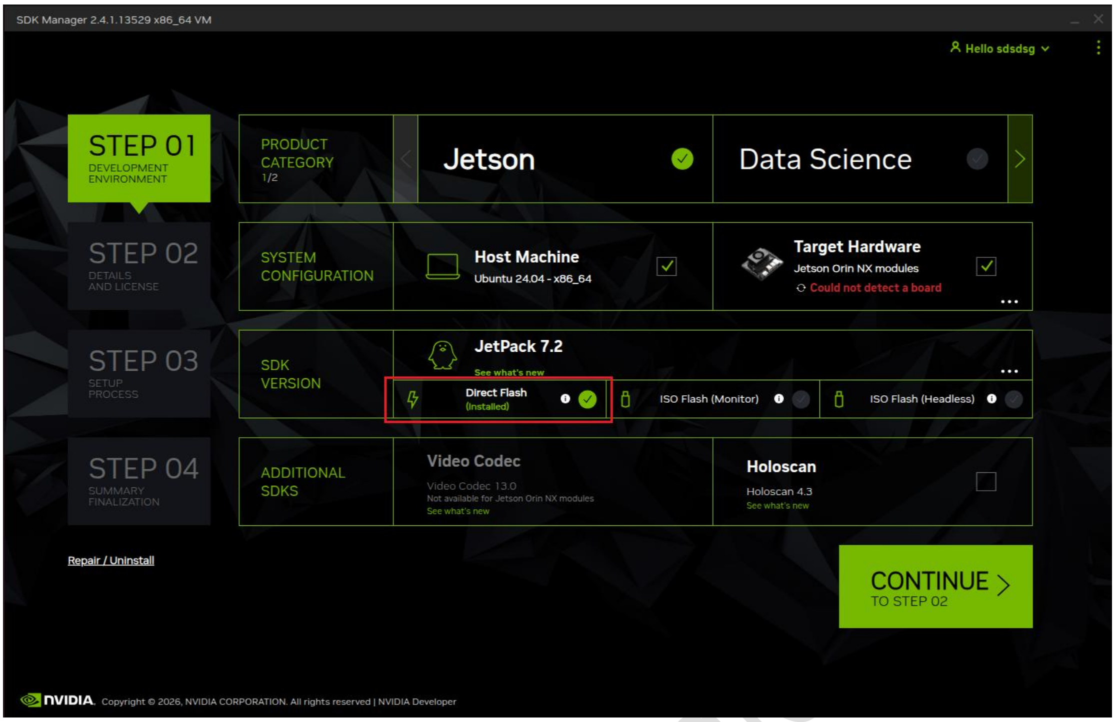
To flash **Jetpack 7.x, an **Ubuntu 24.04** computer is required as the HOST

b) Download the latest **SDK manager** from NVIDIA and install it in **Ubuntu 20.04** (You need to register an NVIDIA account, which will also be used later)

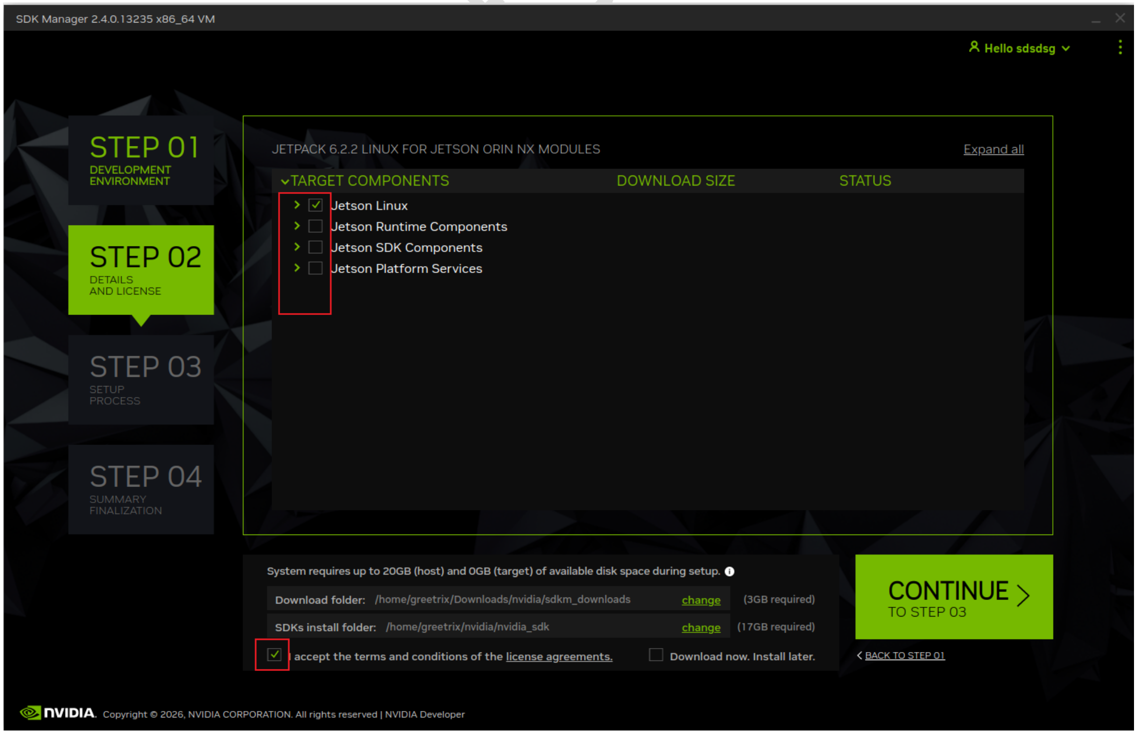
➢ SDK-Manager Download： https://developer.nvidia.com/nvidia-sdk-manager

c) **Continue** Select the required **Target Hardware** and **JetPack** version, **uncheck Host Machine (to save host storage space)**, and check DeepStream/GXF Runtime based on component requirements. Here, **Jetpack 6.2.2 version is selected as an example**, then click **Continue**
***For Jetpack 7.x，Select Direct Flash**



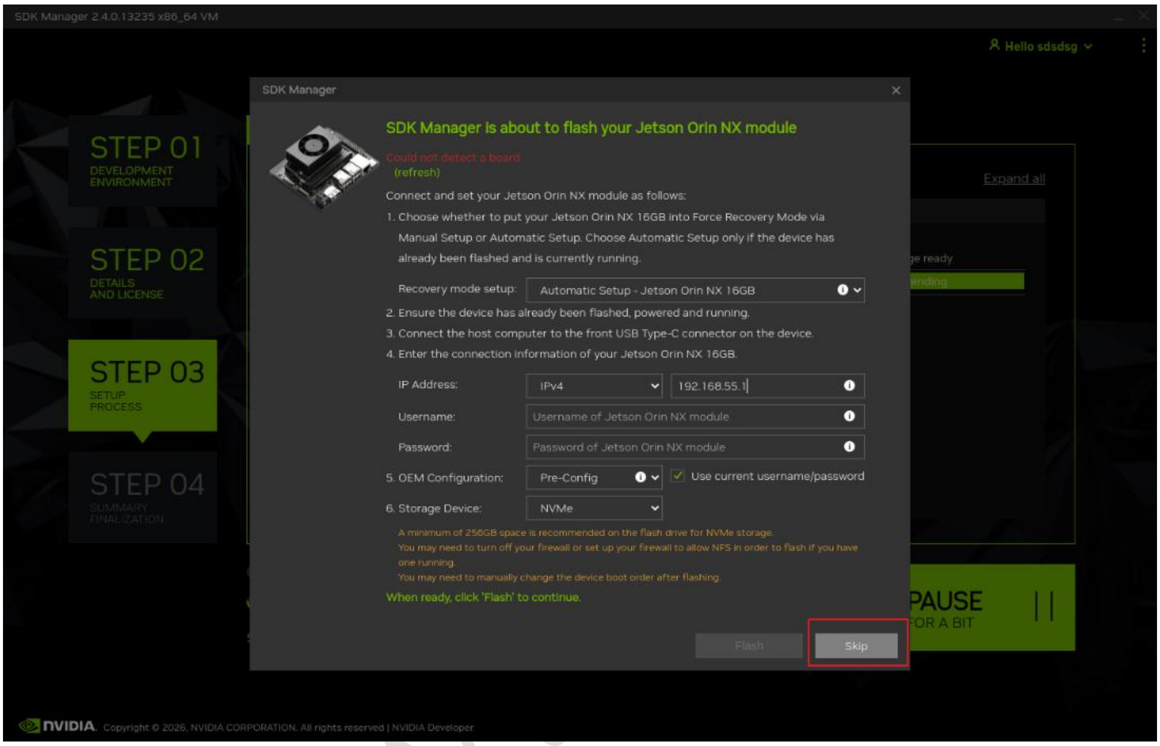


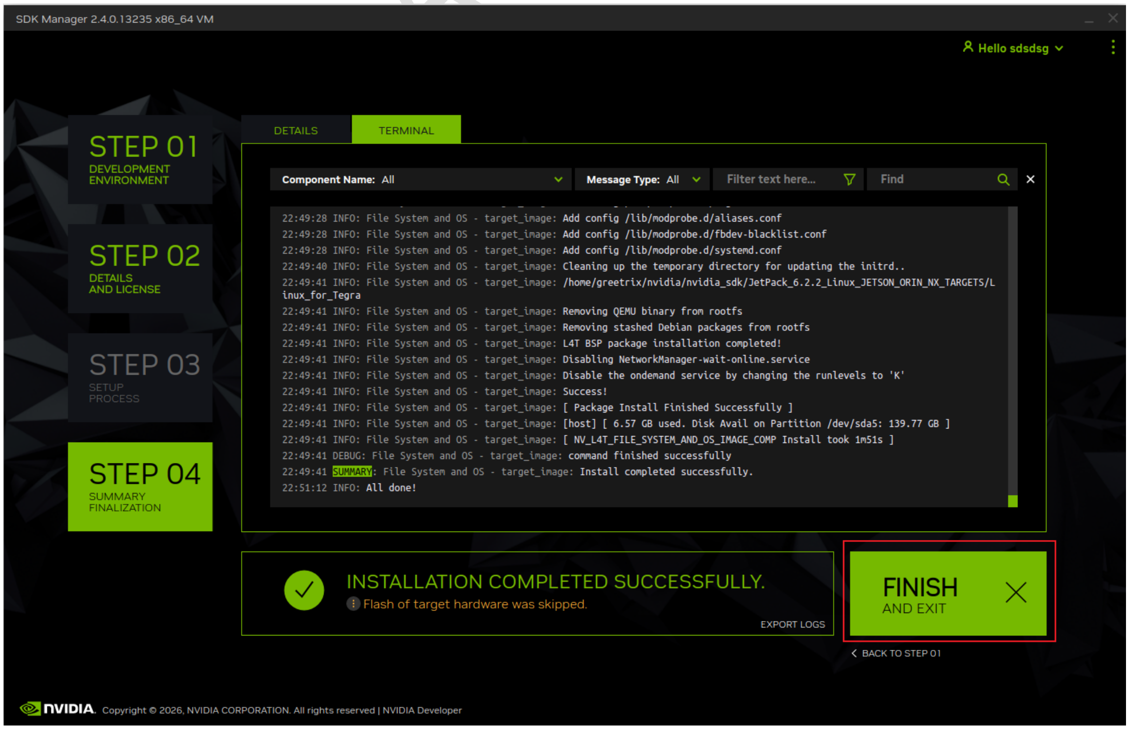
d) Only select **Jetson Linux** here, **uncheck Jetson Runtime/SDK Components/Plat-form Services (the components will be re downloaded and installed together later)**, and click **CONTINUE** to proceed to the next step.



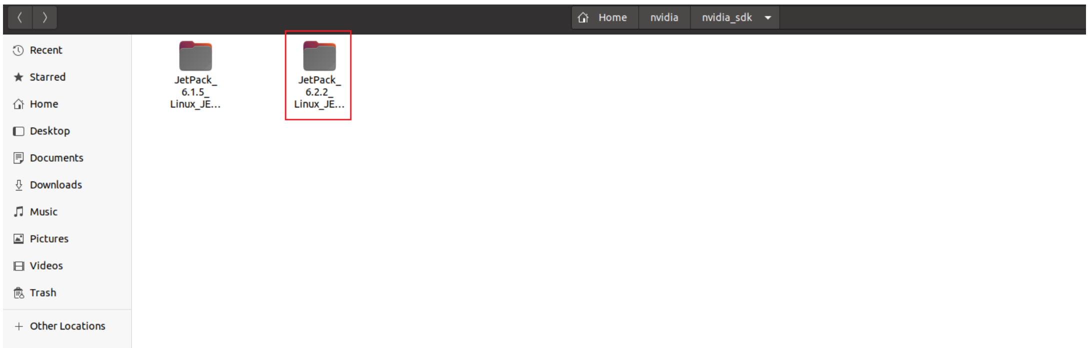
P. S: Please download and install in a smooth network environment. When the download or installation fails, click **Retry** to continue until all the status is installed and green is displayed. During the installation process, a network burning message will pop up and select skip.

(**The subsequent steps involve automatically replacing the device tree and flashing the system via scripts**)



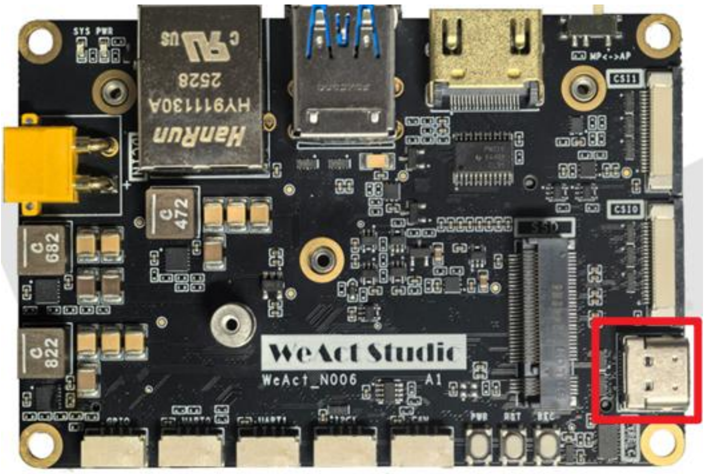


e) After the installation is successful, the required files will be downloaded with the corresponding version under ~/NVIDIA/nvidia_sdk/

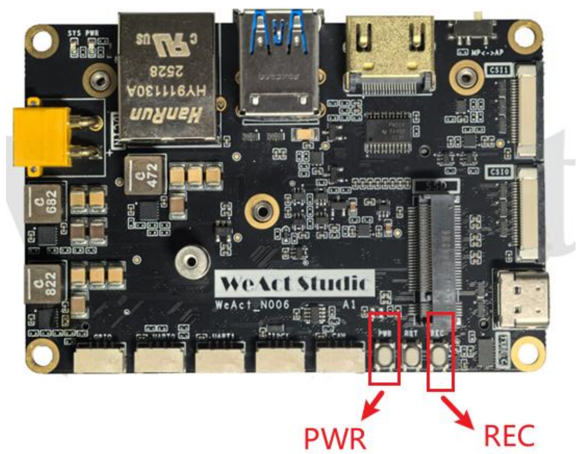


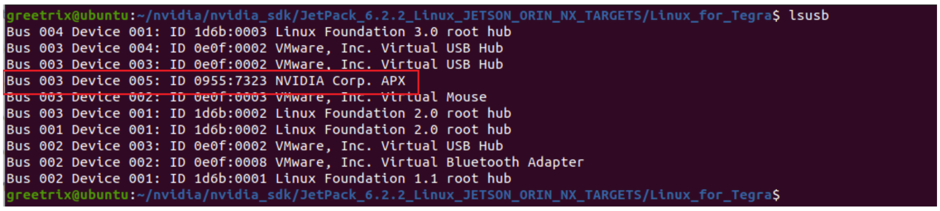
2. FLASH SYSTEM IMAGE

1. Connect **USB Type-C** Cable to carrier board **USB OTG** connector



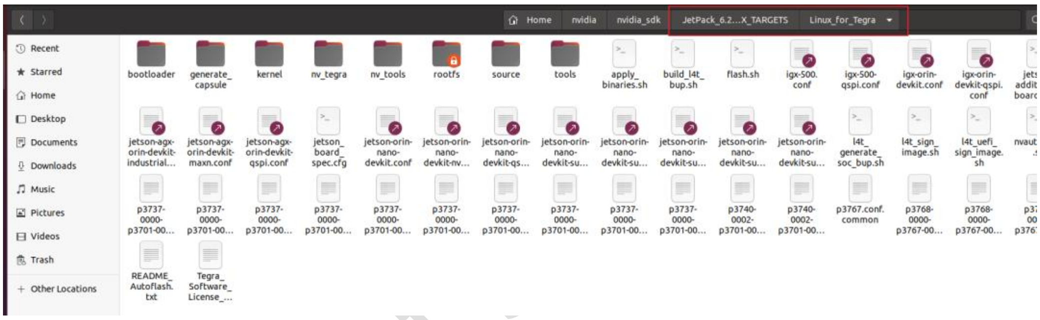
1. Switch slide button to MP**(Manual Power On)**, 2 way enter Recovery mode below
   a) Long press **REC** key, Press **PWR** to power on, Release **REC** key that make the system to enter Recovery mode
   b) Press **PWR** to power on, Long Press **REC** Key, Press **RST** to enter Recovery mode
2. **NVIDIA USB Driver** will appear in the lower right of VMware, or open the terminal and enter **lsusb** command, the NVIDIA Corp will be found and the **fan speed will be set to max.**





*VMwave will display usb logo

3. Enter **~/nvidia/nvidia_sdk/JetPack_<Version>_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra**， <Version>is Jetpack Version Number, Example with 6.2.2, open the terminal：



a) sudo apt-get install git

b) git clone https://github.com/WeActStudio/WeAct_N006

c)

i. sudo bash WeAct_N006/03_Script/weact_n006_orin_flash.sh (Super Mode)

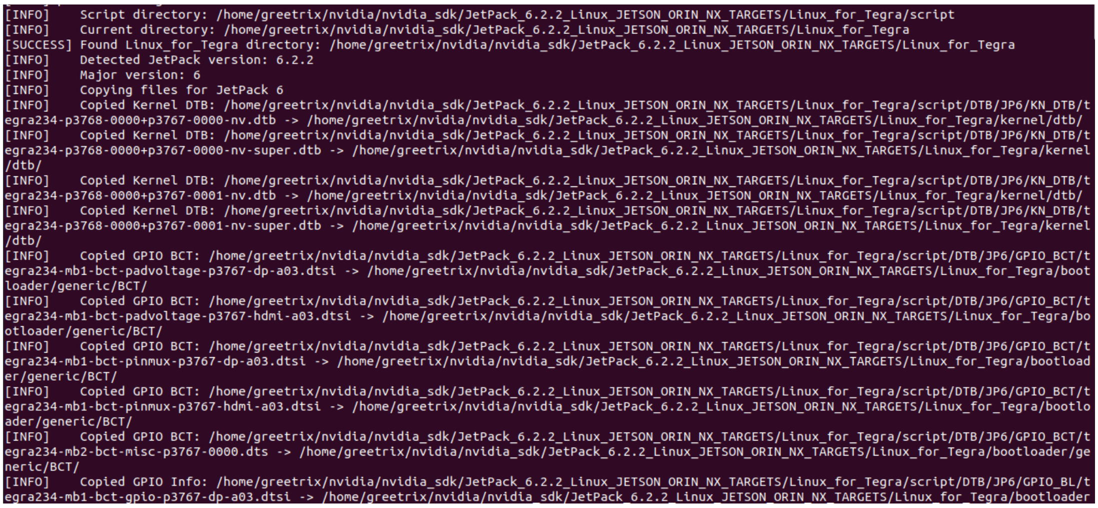
ii. sudo bash WeAct_N006/03_Script/weact_n006_orin_flash.sh -n (Non-Super Mode)

or

b) git clone https://gitee.com/WeAct-TC/weact_n006

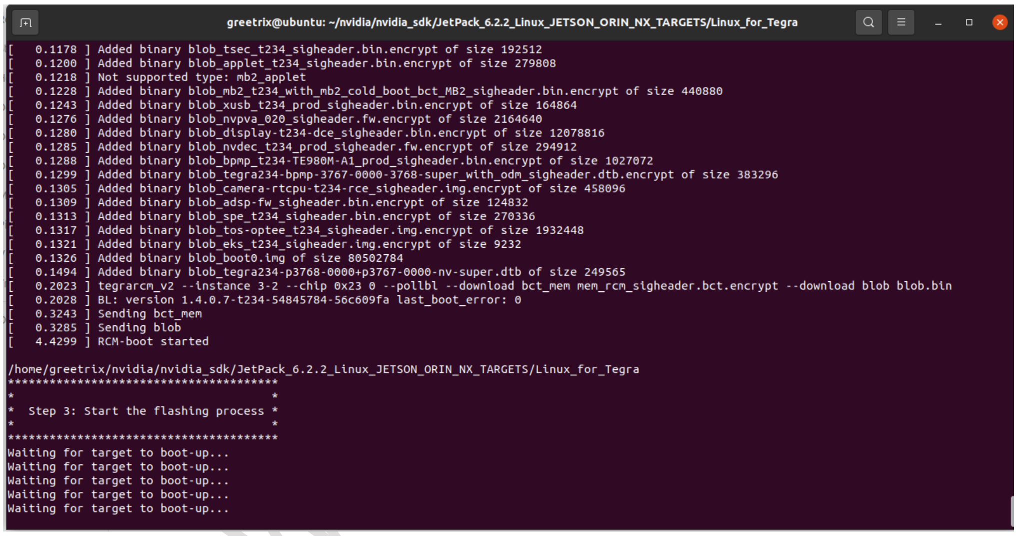
i. c) sudo bash WeAct_N006/03_Script/weact_n006_orin_flash.sh (Super Mode)

ii. sudo bash WeAct_N006/03_Script/weact_n006_orin_flash.sh -n (Non-Super Mode)



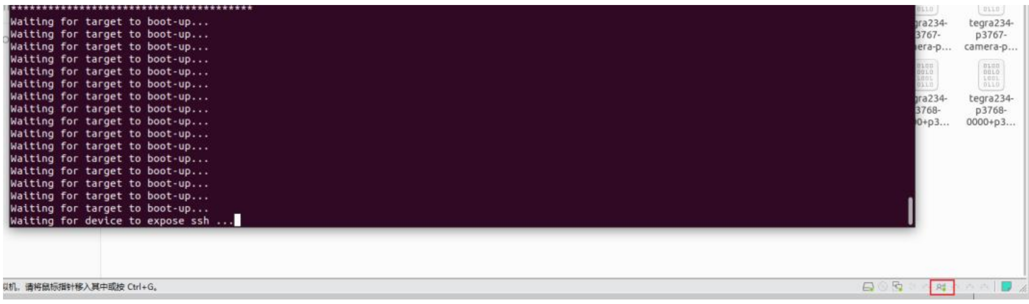
```
[INFO]    Script directory: /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/script
[INFO]    Current directory: /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra
[SUCCESS] Found Linux_for_Tegra directory: /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra
[INFO]    Detected JetPack version: 6.2.2
[INFO]    Major version: 6
[INFO]    Copying files for JetPack 6
[INFO]    Copied Kernel DTB: /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/script/DTB/JP6/KN_DTB/t
egra234-p3768-0000+p3767-0000-nv.dtb -> /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/kernel/dtb/
[INFO]    Copied Kernel DTB: /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/script/DTB/JP6/KN_DTB/t
egra234-p3768-0000+p3767-0000-nv-super.dtb -> /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/kernel
/dtb/
[INFO]    Copied Kernel DTB: /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/script/DTB/JP6/KN_DTB/t
egra234-p3768-0000+p3767-0001-nv.dtb -> /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/kernel/dtb/
[INFO]    Copied Kernel DTB: /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/script/DTB/JP6/KN_DTB/t
egra234-p3768-0000+p3767-0001-nv-super.dtb -> /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/kernel
/dtb/
[INFO]    Copied GPIO BCT: /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/script/DTB/JP6/GPIO_BCT/t
egra234-mb1-bct-padvoltage-p3767-dp-a03.dtsi -> /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/boot
loader/generic/BCT/
[INFO]    Copied GPIO BCT: /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/script/DTB/JP6/GPIO_BCT/t
egra234-mb1-bct-padvoltage-p3767-hdmi-a03.dtsi -> /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/bo
otloader/generic/BCT/
[INFO]    Copied GPIO BCT: /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/script/DTB/JP6/GPIO_BCT/t
egra234-mb1-bct-pinmux-p3767-dp-a03.dtsi -> /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/bootload
er/generic/BCT/
[INFO]    Copied GPIO BCT: /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/script/DTB/JP6/GPIO_BCT/t
egra234-mb1-bct-pinmux-p3767-hdmi-a03.dtsi -> /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/bootlo
ader/generic/BCT/
[INFO]    Copied GPIO BCT: /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/script/DTB/JP6/GPIO_BCT/t
egra234-mb2-bct-misc-p3767-0000.dts -> /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/bootloader/ge
neric/BCT/
[INFO]    Copied GPIO Info: /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/script/DTB/JP6/GPIO_BL/t
egra234-mb1-bct-gpio-p3767-dp-a03.dtsi -> /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/bootloader
```

Script will replace the device-tree and flash the system auto.

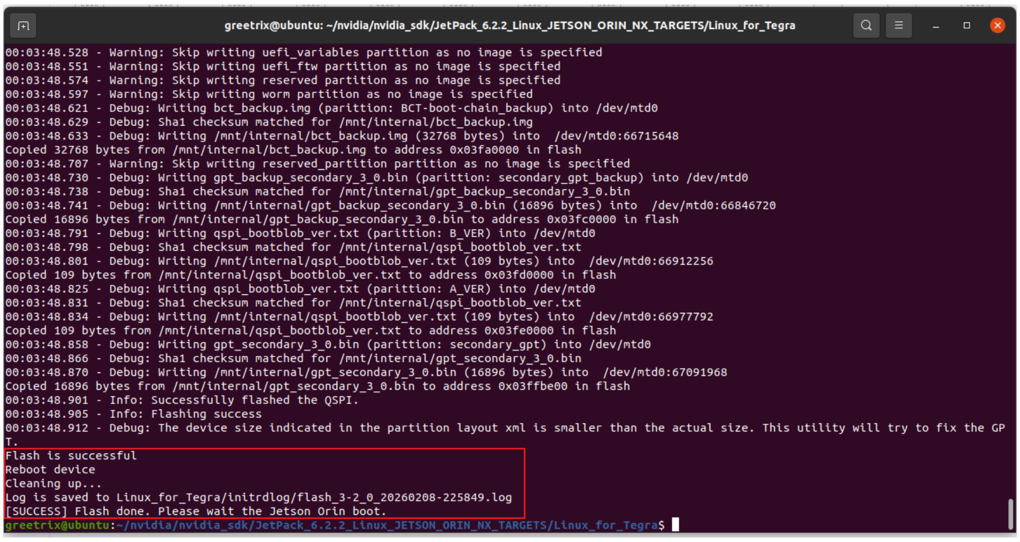


```
greetrix@ubuntu: ~/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra
[   0.1178 ] Added binary blob_tsec_t234_sigheader.bin.encrypt of size 192512
[   0.1200 ] Added binary blob_applet_t234_sigheader.bin.encrypt of size 279808
[   0.1218 ] Not supported type: mb2_applet
[   0.1228 ] Added binary blob_mb2_t234_with_mb2_cold_boot_bct_MB2_sigheader.bin.encrypt of size 440880
[   0.1243 ] Added binary blob_xusb_t234_prod_sigheader.bin.encrypt of size 164864
[   0.1276 ] Added binary blob_nvpva_020_sigheader.fw.encrypt of size 2164640
[   0.1280 ] Added binary blob_display-t234-dce_sigheader.bin.encrypt of size 12078816
[   0.1285 ] Added binary blob_nvdec_t234_prod_sigheader.fw.encrypt of size 294912
[   0.1288 ] Added binary blob_bpmp_t234-TE980M-A1_prod_sigheader.bin.encrypt of size 1027072
[   0.1299 ] Added binary blob_tegra234-bpmp-3767-0000-3768-super_with_odm_sigheader.dtb.encrypt of size 383296
[   0.1305 ] Added binary blob_camera-rtcpu-t234-rce_sigheader.img.encrypt of size 458096
[   0.1309 ] Added binary blob_adsp-fw_sigheader.bin.encrypt of size 124832
[   0.1313 ] Added binary blob_spe_t234_sigheader.bin.encrypt of size 270336
[   0.1317 ] Added binary blob_tos-optee_t234_sigheader.img.encrypt of size 1932448
[   0.1321 ] Added binary blob_eks_t234_sigheader.img.encrypt of size 9232
[   0.1326 ] Added binary blob_boot0.img of size 80502784
[   0.1494 ] Added binary blob_tegra234-p3768-0000+p3767-0000-nv-super.dtb of size 249565
[   0.2023 ] tegrarcm_v2 --instance 3-2 --chip 0x23 0 --pollbl --download bct_mem mem_rcm_sigheader.bct.encrypt --download blob blob.bin
[   0.2028 ] BL: version 1.4.0.7-t234-54845784-56c609fa last_boot_error: 0
[   0.3243 ] Sending bct_mem
[   0.3285 ] Sending blob
[   4.4299 ] RCM-boot started

/home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra
***************************************
*                                     *
*  Step 3: Start the flashing process *
*                                     *
***************************************
Waiting for target to boot-up...
Waiting for target to boot-up...
Waiting for target to boot-up...
Waiting for target to boot-up...
Waiting for target to boot-up...
```

Not USB will re-connect. If it is a virtual machine, please grant the permission for automatic USB connection in advance



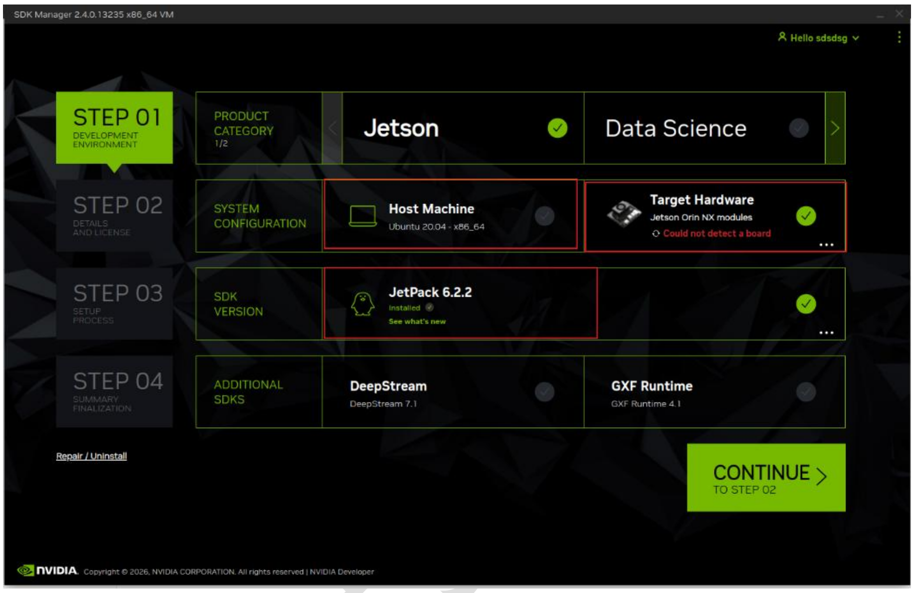
```
***************************************
Waiting for target to boot-up...
Waiting for target to boot-up...
Waiting for target to boot-up...
Waiting for target to boot-up...
Waiting for target to boot-up...
Waiting for target to boot-up...
Waiting for target to boot-up...
Waiting for target to boot-up...
Waiting for target to boot-up...
Waiting for target to boot-up...
Waiting for target to boot-up...
Waiting for target to boot-up...
Waiting for target to boot-up...
Waiting for target to boot-up...
Waiting for target to boot-up...
Waiting for target to boot-up...
Waiting for device to expose ssh ...
```



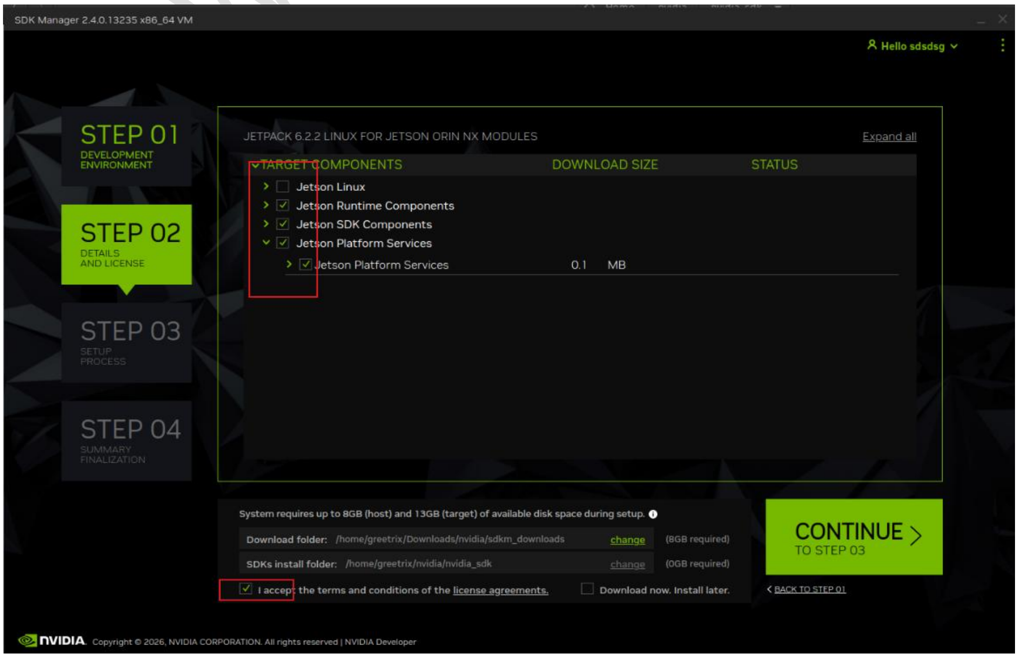
After updating the device tree, it will be successful! Display, as shown in the following figure

3. INSTALL NVIDIA COMPONENT

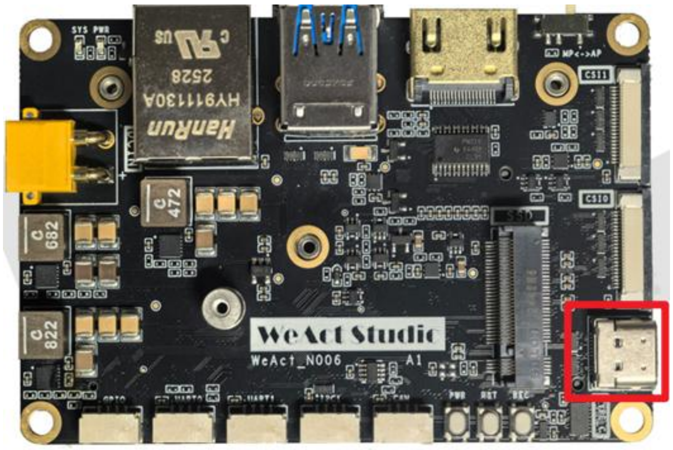
1. Select the target **hardware** and **jetpack** version required, uncheck the host machine, and click **CONTINUE**.



2. Check the required **SDK components**, check **I accept the terms and conditions of the license agreements,** and click **CONTINUE** to proceed to the next step.

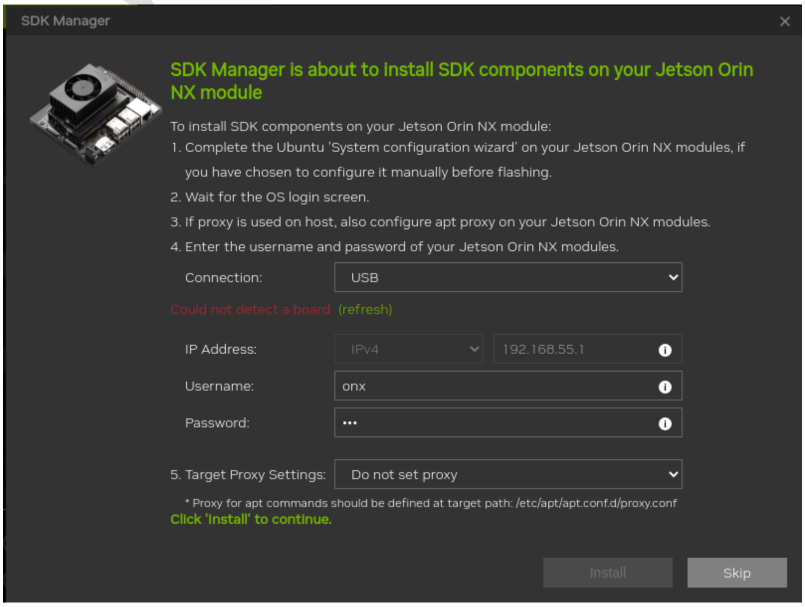


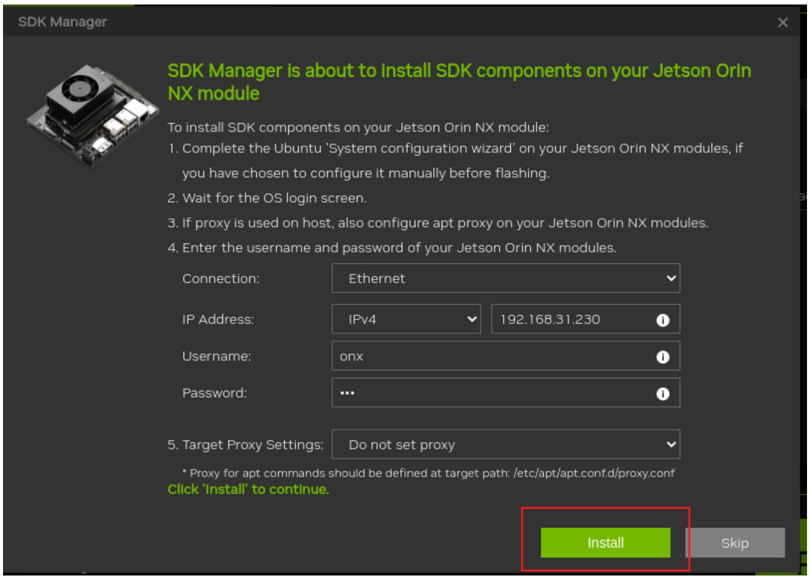
3. Use the **USB type-C cable** to connect the **USB OTG interface** on the carrier board.



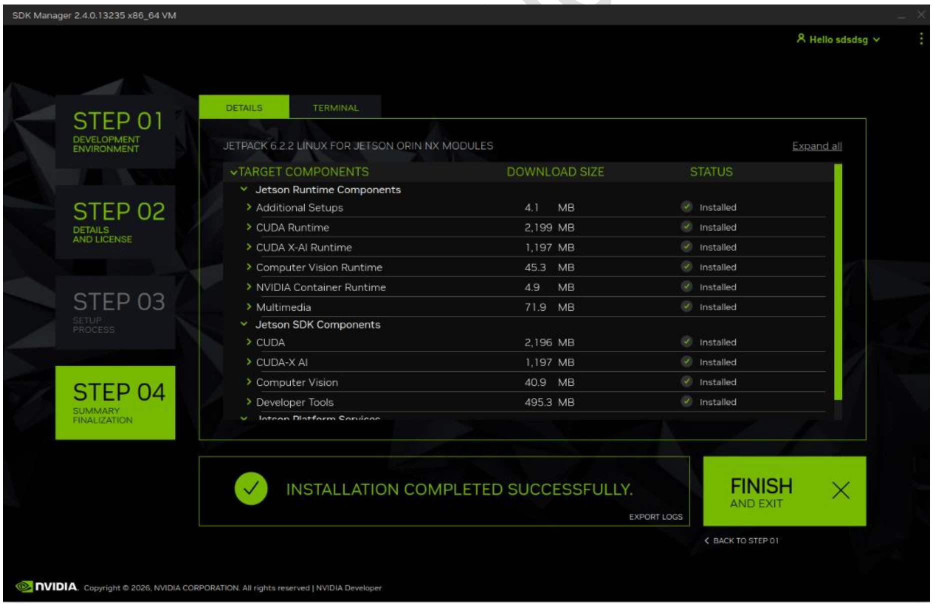
4. Turn the **power on key to MP (manual power on)** and **press PWR key** to power on. At this time, **NVIDIA USB drive sign** will appear in the lower right corner of VMware, or open the terminal and enter **lsusb** command to find **NVIDIA Corp**.

5. Enter the **Orin NX** account password, and ensure the **Orin NX device remains connected to the internet**. Alternatively, you can use the IP address within the same local network.(**Figure 2: Host and ORIN NX connected to the same router**)



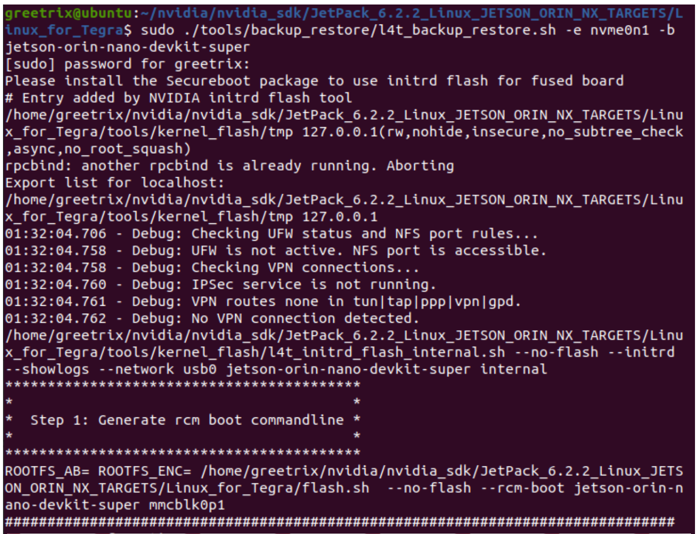


6. Wait for the installation to complete.

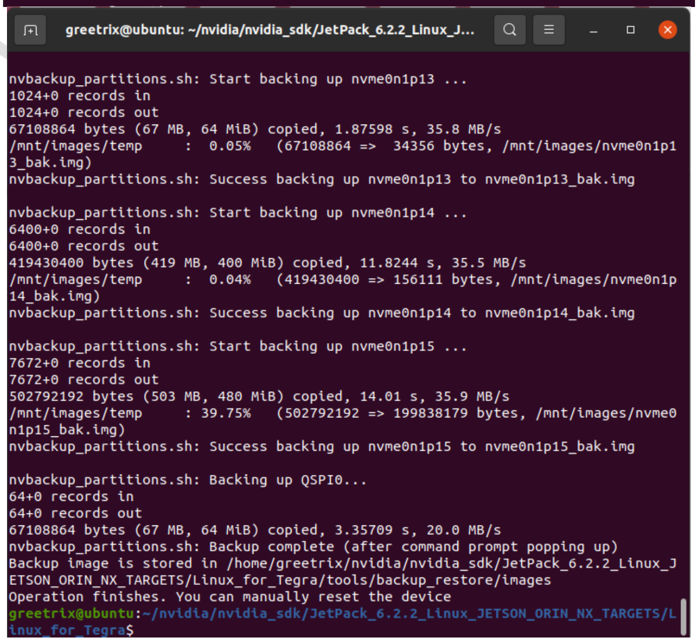


4. BACKUP AND RECOVERY SYSTEM

1. Use a flashing environment that matches the system Jetpack version, execute the following command, and wait for the backup to complete
   a) sudo ./tools/backup_restore/l4t_backup_restore.sh -e nvme0n1 -b jetson-orin-nano-devkit-super (Super Mode)
   b) sudo ./tools/backup_restore/l4t_backup_restore.sh -e nvme0n1 -b jetson-orin-nano-devkit (Non-Super Mode)

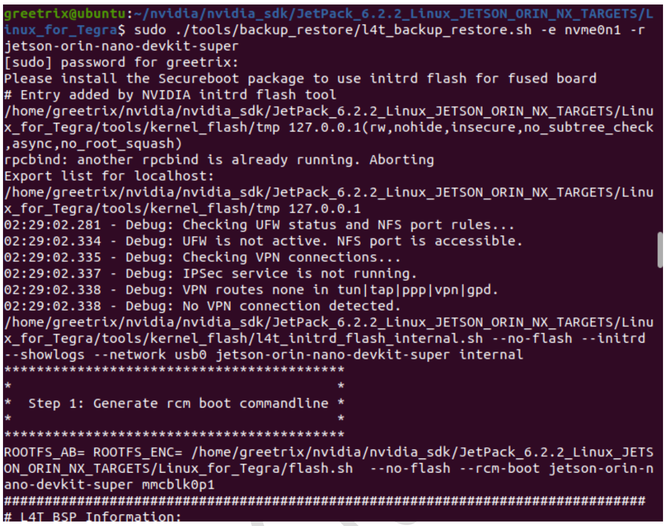


```
greetrix@ubuntu:~/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra$ sudo ./tools/backup_restore/l4t_backup_restore.sh -e nvme0n1 -b jetson-orin-nano-devkit-super
[sudo] password for greetrix:
Please install the Secureboot package to use initrd flash for fused board
# Entry added by NVIDIA initrd flash tool
/home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/tools/kernel_flash/tmp 127.0.0.1(rw,nohide,insecure,no_subtree_check,async,no_root_squash)
rpcbind: another rpcbind is already running. Aborting
Export list for localhost:
/home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/tools/kernel_flash/tmp 127.0.0.1
01:32:04.706 - Debug: Checking UFW status and NFS port rules...
01:32:04.758 - Debug: UFW is not active. NFS port is accessible.
01:32:04.758 - Debug: Checking VPN connections...
01:32:04.760 - Debug: IPSec service is not running.
01:32:04.761 - Debug: VPN routes none in tun|tap|ppp|vpn|gpd.
01:32:04.762 - Debug: No VPN connection detected.
/home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/tools/kernel_flash/l4t_initrd_flash_internal.sh --no-flash --initrd --showlogs --network usb0 jetson-orin-nano-devkit-super internal
*******************************************
*                                         *
*  Step 1: Generate rcm boot commandline *
*                                         *
*******************************************
ROOTFS_AB= ROOTFS_ENC= /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/flash.sh  --no-flash --rcm-boot jetson-orin-nano-devkit-super mmcblk0p1
################################################################################
```



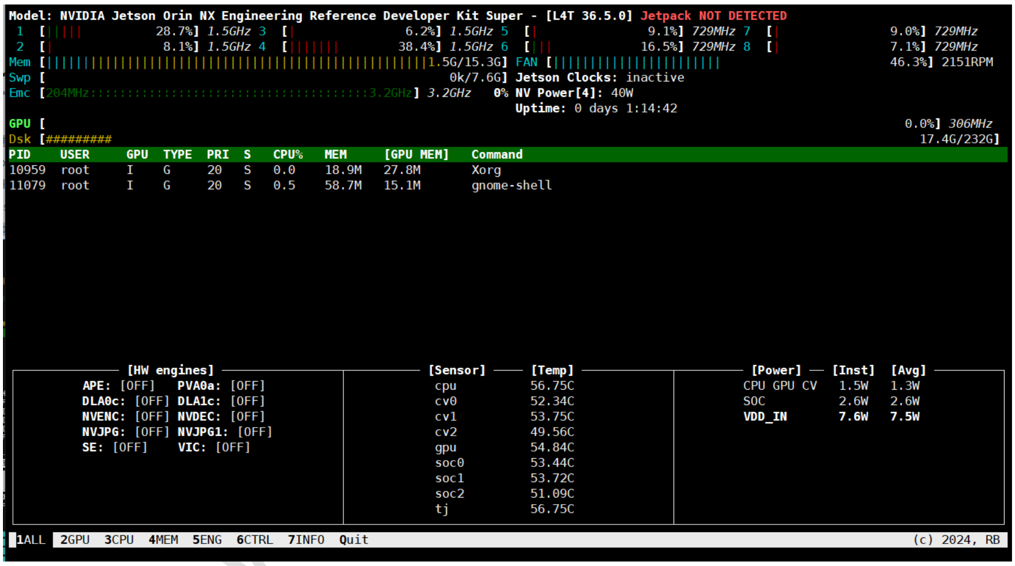
```
greetrix@ubuntu: ~/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_J...

nvbackup_partitions.sh: Start backing up nvme0n1p13 ...
1024+0 records in
1024+0 records out
67108864 bytes (67 MB, 64 MiB) copied, 1.87598 s, 35.8 MB/s
/mnt/images/temp     :  0.05%   (67108864 =>  34356 bytes, /mnt/images/nvme0n1p13_bak.img)
nvbackup_partitions.sh: Success backing up nvme0n1p13 to nvme0n1p13_bak.img

nvbackup_partitions.sh: Start backing up nvme0n1p14 ...
6400+0 records in
6400+0 records out
419430400 bytes (419 MB, 400 MiB) copied, 11.8244 s, 35.5 MB/s
/mnt/images/temp     :  0.04%   (419430400 => 156111 bytes, /mnt/images/nvme0n1p14_bak.img)
nvbackup_partitions.sh: Success backing up nvme0n1p14 to nvme0n1p14_bak.img

nvbackup_partitions.sh: Start backing up nvme0n1p15 ...
7672+0 records in
7672+0 records out
502792192 bytes (503 MB, 480 MiB) copied, 14.01 s, 35.9 MB/s
/mnt/images/temp     : 39.75%   (502792192 => 199838179 bytes, /mnt/images/nvme0n1p15_bak.img)
nvbackup_partitions.sh: Success backing up nvme0n1p15 to nvme0n1p15_bak.img

nvbackup_partitions.sh: Backing up QSPI0...
64+0 records in
64+0 records out
67108864 bytes (67 MB, 64 MiB) copied, 3.35709 s, 20.0 MB/s
nvbackup_partitions.sh: Backup complete (after command prompt popping up)
Backup image is stored in /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/tools/backup_restore/images
Operation finishes. You can manually reset the device
greetrix@ubuntu:~/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra$
```

2. In the backup burning environment, use the following command
    a) sudo ./tools/backup_restore/l4t_backup_restore.sh -e nvme0n1 -r jetson-orin-nano-devkit-super (Super Mode)
    b) sudo ./tools/backup_restore/l4t_backup_restore.sh -e nvme0n1 -r jetson-orin-nano-devkit (Non-Super Mode)

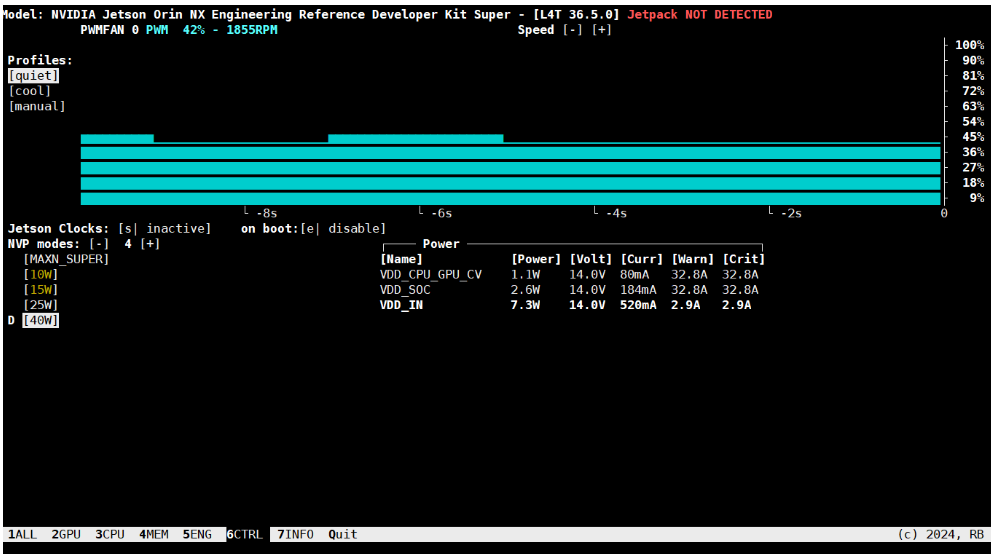


5. INSTALL JTOP(CONTROL FANS/GET SYSTEM INFO)

1. Open terminal, use below command install JTOP
   a) sudo apt-get install python3-pip python3-dev -y
   b) sudo -H pip3 install jetson-stats --break-system-packages
   c) sudo systemctl restart jtop.service
   d) sudo jtop



2. Click on 6CTRL to manually control the fan speed and power scheme.



6. USING CAN FOR COMMUNICATION

1. OrinNX ntegrates one CAN controller CAN0, and WeAct Studio has designed one CAN transceiver (CAN0) on the carrier board, which can **be directly mounted on the CAN physical bus** for use.

2. OrinNX comes with a canbus driver and is integrated into the image, supporting canbus without further processing. We need to install the canbus module. (Enter the following command on the terminal or put it into rc.local to enable self start)

   sudo modprobe can
   sudo modprobe can-raw
   sudo modprobe can-bcm
   sudo modprobe can-gw
   sudo modprobe can_dev
   sudo modprobe mttcan

3. Check if the installation was successful through **lsmod**.

```
root@onx:/home/onx# lsmod | grep can
can_gw                 24576  0
can_bcm                32768  0
can_raw                28672  0
can                    28672  3 can_gw,can_raw,can_bcm
mttcan                 65536  0
nvpps                  28672  1 mttcan
can_dev                36864  1 mttcan
root@onx:/home/onx#
```

4. Configure canbus properties, similar to the baud rate setting of the serial port.

   sudo ip link set can0 type can bitrate 500000 sudo ip link set up can0

5. Check if the configuration was successful through ifconfig.

```
root@onx:/home/onx# sudo ip link set can0 type can bitrate 500000
root@onx:/home/onx# sudo ip link set up can0
root@onx:/home/onx# ifconfig
can0: flags=193<UP,RUNNING,NOARP>  mtu 16
        unspec 00-00-00-00-00-00-00-00-00-00-00-00-00-00-00-00  txqueuelen 10  (UNSPEC)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
        device interrupt 201
```

6. Through cansend can0 (can1) on a terminal ××× Command to send data, and another terminal completes the actual signal transmission and reception test through candump can1 (can0)


```
root@onx:/home/onx# candump can0
  can0  555   [2]  AA AA
  can0  555   [2]  AA AA
  can0  555   [2]  AA AA
  can0  555   [2]  AA AA
  can0  555   [2]  AA AA
  can0  555   [2]  AA AA
```

```
root@onx:/home/onx# cansend can0 555#AAAA
root@onx:/home/onx# cansend can0 555#AAAA
root@onx:/home/onx# cansend can0 555#AAAA
root@onx:/home/onx# cansend can0 555#AAAA
root@onx:/home/onx# cansend can0 555#AAAA
root@onx:/home/onx# cansend can0 555#AAAA
root@onx:/home/onx#
```

7. GPIO USAGE IN SHELL

1. Jetson OrinNX can control the GPIO input or output through shell command, install gpiod by below command.
   a) sudo apt-get install gpiod
   b) sudo apt-get install libgpiod-dev

2. GPIO Usage

	Orin NX 8G/16G
GPIO1	PG.06
GPIO2	PAC.06
GPIO3	PN.01
GPIO4	PH.00

   a. GPIO Output: Example with GPIO1
      i. gpioset `gpiofind "PG.06"`=0 (Set Output Low Level)
      ii. gpioset `gpiofind "PG.06"`=1 (Set Output High Level)
   b. GPIO Input: Example with GPIO1
      i. gpioget `gpiofind "PG.06"`

8. UART/I2C/CSI USAGE

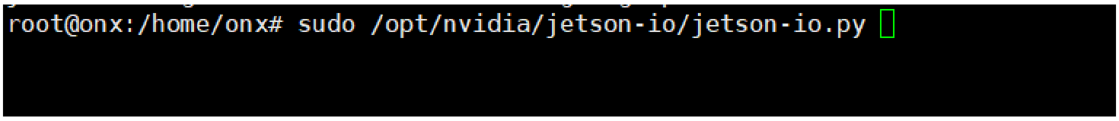
1. UART Usage

	Jetpack 5.X	Jetpack 6.X/7.X
UART0	ttyTHS1	ttyTHS0
UART1	ttyTHS0	ttyTHS1

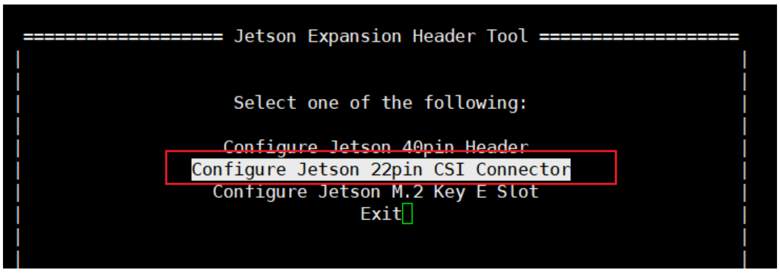
a. Install cutecom/minicom to use UART by below command.
   a) sudo apt-get install cutecom
   b) sudo apt-get install minicom

2. **I2C Usage** (I2C in carrier board is I2C@7), you can get the I2C device address by below command.
   a) sudo i2cdetect -r -y 7

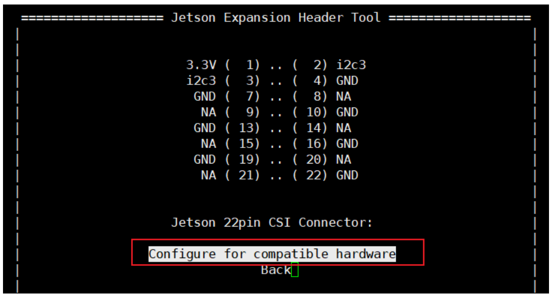
3. **CSI Usage** (Example with dual IMX-219)
   a) Command: **sudo /opt/nvidia/jetson-io/jetson-io.py**



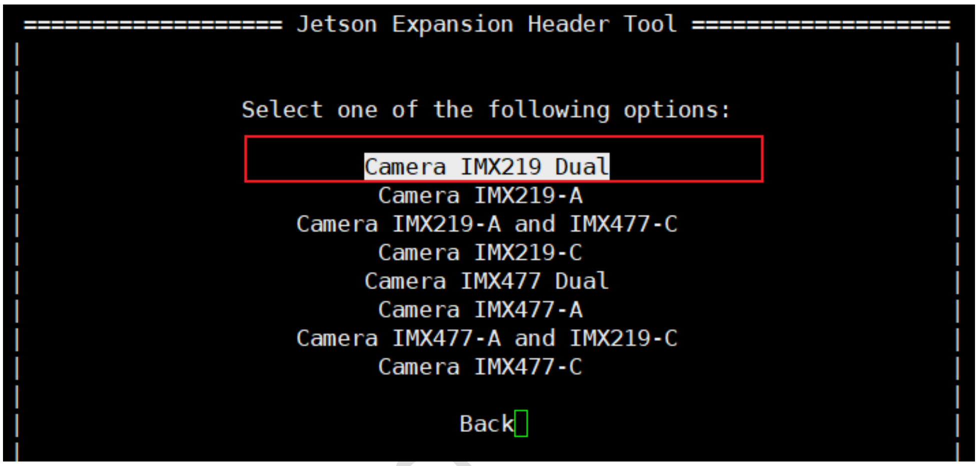
   b) Select **Configure Jetson 22pin CSI Connector.**



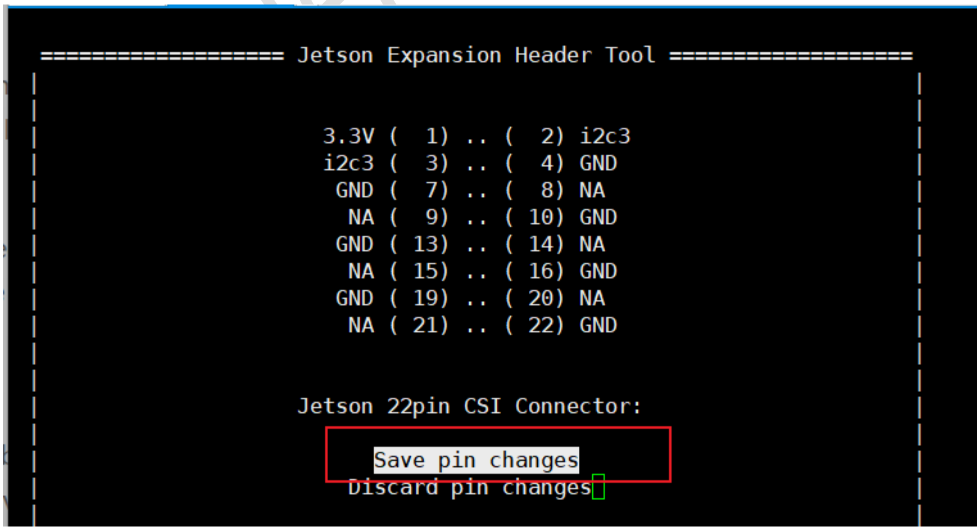
   c) Select **Configure for compatible hardware.**



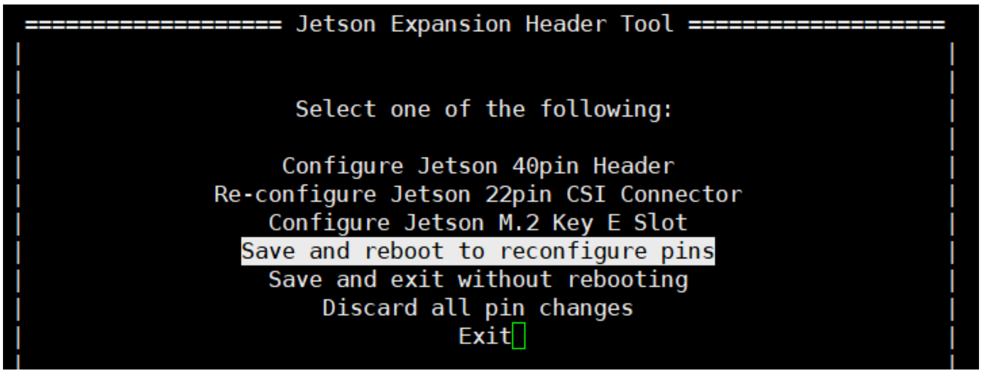
d) Select **Camera IMX219 Dual.**



e) Select **Save pin changes.**



f) Select **Save and reboot to reconfigure pins.**



g) Wait to re-boot and camera can be used.

CONTACT US

1. Github：https://github.com/WeActStudio
2. Gitee：https://gitee.com/WeAct-TC
3. Taobao：https://weactstudio.taobao.com/
4. Ali-express: Search WeAct in Ali-express